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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2024

END SEMESTER EXAMINATION - EVEN 2024
HU204 - ENVIRONMENTAL SCIENCE

TIME ALLOTTED: 3Hours

FULL MARKS: 70

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A
(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

- 1.i) Identify the prime component of London smog (1) CO1 BL1
- a) water
- b) sulphur dioxide
- c) carbon dioxide
- d) carbon monooxide
- 1.ii) Black foot disease occurs due to (1) CO3 BL1
- a) Cd toxicity
- b) Pb toxicity
- c) As toxicity
- d) Hg toxicity
- 1.iii) The unit of Sound intensity is (1) CO4 BL1
- a) candela
- b) Watt/m²
- c) decibel
- d) All of these
- 1.iv) Ozone is a pollutant when present in (1) CO3 BL1
- a) stratosphere
- b) troposphere
- c) mesosphere
- d) ionosphere
- 1.v) Which gas is removed by the help of a scrubber? (1) CO1 BL4
- a) NO₂
- b) CO₂
- c) NH₃

d) SO₂

1.vi) The example of secondary consumer (1) CO₂ BL1

a) Plant

b) Cow

c) Tiger

d) Frog

1.vii) Which of the following is not biodegradable? (1) CO₂ BL1

a) polyethylene

b) cotton

c) vegetable waste

d) wood

1.viii) Lakes rich in nutrients is known as (1) CO₂ BL2

a) Eutrophic lake

b) Mesotrophic lake

c) Atrophic lake

d) Oligotrophic lake

1.ix) Blue baby syndrome is related to (1) CO₂ BL2

a) nitrate

b) nitrite

c) sulphate

d) phosphate

1.x) Food Web is (1) CO₁ BL1

a) Single food chain

b) Disintegrated food chain

c) Interconnected food chain

d) None of these

1.xi) Example of non renewable resource (1) CO₁ BL1

a) Solar Energy

b) Wind energy

c) Fossil fuel

d) Tidal Energy

1.xii) Which of the following causes disinfection of water (1) CO₃ BL2

a) Chlorine

b) Ozone

c) Hydrogen

d) Both a and b

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

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| 2. Answer the following questions: | (5) | | |
| a) What is acid rain? | (1) | CO1 | BL2 |
| b) Write down the causes, reactions and effects of acid rain. | (4) | CO1 | BL2 |
| 3. Write a Short note on Pond Eco system. | (5) | CO3 | BL2 |
| 4. Write a short note on global warming and green house gases. | (5) | CO4 | BL3 |
| 5. Calculate the intensity of 100 dB sound (Reference intensity : 10-12 w/m ²) | (5) | CO3 | BL2 |
| 6. Classify solid wastes with examples | (5) | CO3 | BL2 |

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

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| 7. Answer the following | (15) | | |
| a) A work area is composed of 95 dB sound level for 1 hr., 80 dB for 3hrs and 75 dB for 4hrs. Maximum permissible limits for 95 dB is 2 hrs per day, for 80 db is 16 hrs per day and infinite time for 75 dB. Predict whether the work area is suitable for health or not. | (3) | CO3 | BL3 |
| b) A 10 mL of sewage water mixed with enough water to fill a 300 mL bottle has an initial DO of 9.0 mg/L. To help assure an accurate test, it is desirable to have at least a 2.0 mg/L drop in DO during the five day run, and the final DO should be at least 2.0 mg/L. For what range of BOD5 would this dilution produce the desired result? | (5) | CO3 | BL3 |
| c) What are the essential characteristics of hazardous waste? | (2) | CO2 | BL1 |
| d) Distinguish between Photochemical smog and London smog. | (5) | CO4 | BL1 |
| 8a. Define Eutrophication. How can we control eutrophication? | (5) | CO3 | BL2 |
| 8b. What is Environmental Impact Assessment ? | (3) | CO3 | BL2 |
| 8c. What are the differences between food chain & food web | (2) | CO3 | BL2 |
| 8d. Write a short note on grassland eco system | (5) | CO3 | BL2 |
| 9a. Mention five criteria air pollutant along with biochemical effects. | (5) | CO3 | BL2 |
| 9b. Classify Noise along with example. | (5) | CO3 | BL2 |
| 9c. Write a short Note on Renewable and Non-renewable Resources | (5) | CO4 | BL2 |
| 10. Answer the following questions: | (15) | | |
| a) Explain the 5 day BOD test. | (5) | CO3 | BL3 |
| b) Describe surface water treatment in brief along with schematic diagram. | (5) | CO1 | BL2 |

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| c) How can a cyclone separator and catalytic converter be used to control the air quality? | (5) | CO2 | BL2 |
| 11a. Define Ozone layer depletion with reaction. | (5) | CO3 | BL2 |
| 11b. Answer the following questions: | (5) | | |
| a) Mention two major effects of noise pollution. | (2) | CO3 | BL3 |
| b) How can noise be controlled? | (3) | CO3 | BL4 |
| 11c. The increase in population from 1 million to 10 million took 200 years. For exponential growth at constant rate , find out the growth rate(in percentage) | (3) | CO3 | BL2 |
| 11d. What are the advantages of Landfilling? | (2) | CO2 | BL2 |
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